Real-time estimation of the wall orientation and the motion gains

2026-10-19 three axes; $\bar{a}'$ as a state variable; measurement coordinate (x, y, z), local coordinate (u, v, w)

1 Motion gain matrix as a function of the wall orientation

$$\mathbf{V} = \mathbf{R}_y(\theta_y)\,\mathbf{R}_x(\theta_x) = [\,\mathbf{u}\;\;\mathbf{v}\;\;\mathbf{w}\,]$$

$$\bar{\mathbf{\Lambda}} = \mathrm{diag}(\bar{a}_T,\ \bar{a}_T,\ \bar{a}_N)$$

$$\bar{\mathbf{A}} = \mathbf{V}\,\bar{\mathbf{\Lambda}}\,\mathbf{V}^T = \bar{a}_T\,\mathbf{I} + (\bar{a}_N - \bar{a}_T)\,\mathbf{w}\,\mathbf{w}^T$$

$$\mathbf{M}[k] = \Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,\delta\bar{\mathbf{p}}_3[k]$$

2 State of the process

Motion model, control law and dynamics of tracking error (three axes)

$$\bar{\mathbf{p}}[k+1] = \bar{\mathbf{p}}[k] + \bar{\mathbf{A}}[k]\left(\bar{\mathbf{f}}_d[k] + \bar{\mathbf{f}}_T[k]\right)$$

$$\delta\bar{\mathbf{p}}[k] = \bar{\mathbf{p}}_d[k] - \bar{\mathbf{p}}[k]$$

$$\bar{\mathbf{p}}_m[k] = \bar{\mathbf{p}}[k-d] + \mathbf{n}_p[k]$$

$$\delta\bar{\mathbf{p}}_m[k] = \bar{\mathbf{p}}_d[k-d] - \bar{\mathbf{p}}_m[k]$$

$$\hat{\bar{\mathbf{p}}}[k] = \bar{\mathbf{p}}_m[k] + \sum_{i=1}^{d} \hat{\bar{\mathbf{A}}}[k-i]\,\bar{\mathbf{f}}_d[k-i]$$

$$\bar{\mathbf{f}}_d[k] = \hat{\bar{\mathbf{A}}}^{-1}[k]\left\{\bar{\mathbf{p}}_d[k+1] - \hat{\bar{\mathbf{p}}}[k] - \lambda_c\left(\bar{\mathbf{p}}_d[k] - \hat{\bar{\mathbf{p}}}[k]\right)\right\}$$

$$\delta\bar{\mathbf{p}}[k+1] = \lambda_c\,\delta\bar{\mathbf{p}}[k] - \left(\bar{\mathbf{A}}[k] - \hat{\bar{\mathbf{A}}}[k]\right)\mathbf{F}_{dp}[k] - \boldsymbol{\varepsilon}_p[k]$$

$$\mathbf{F}_{dp}[k] = \bar{\mathbf{f}}_d[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{\mathbf{f}}_d[k-i]$$

$$\boldsymbol{\varepsilon}_p[k] = \bar{\mathbf{A}}[k]\,\bar{\mathbf{f}}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{\mathbf{A}}[k-i]\,\bar{\mathbf{f}}_T[k-i] - (1-\lambda_c)\,\mathbf{n}_p[k]$$

True state of the process

$$\delta\bar{\mathbf{p}}_1[k+1] = \delta\bar{\mathbf{p}}_2[k]$$

$$\delta\bar{\mathbf{p}}_2[k+1] = \delta\bar{\mathbf{p}}_3[k]$$

$$\delta\bar{\mathbf{p}}_3[k+1] = \lambda_c\,\delta\bar{\mathbf{p}}_3[k] - \left(\bar{\mathbf{A}}[k] - \hat{\bar{\mathbf{A}}}[k]\right)\mathbf{F}_{dp}[k] - \boldsymbol{\varepsilon}_p[k]$$

$$\bar{a}_T[k+1] = \bar{a}_T[k] + \bar{a}'_T[k]\,\mathbf{w}^T\left(\mathbf{M}[k] + \left(\bar{\mathbf{A}}[k] - \hat{\bar{\mathbf{A}}}[k]\right)\mathbf{F}_{dp}[k] + \boldsymbol{\varepsilon}_p[k]\right)$$

$$\bar{a}'_T[k+1] = \bar{a}'_T[k] + w_{\bar{a}'_T}[k]$$

$$\bar{a}_N[k+1] = \bar{a}_N[k] + \bar{a}'_N[k]\,\mathbf{w}^T\left(\mathbf{M}[k] + \left(\bar{\mathbf{A}}[k] - \hat{\bar{\mathbf{A}}}[k]\right)\mathbf{F}_{dp}[k] + \boldsymbol{\varepsilon}_p[k]\right)$$

$$\bar{a}'_N[k+1] = \bar{a}'_N[k] + w_{\bar{a}'_N}[k]$$

$$\theta_x[k+1] = \theta_x[k]$$

$$\theta_y[k+1] = \theta_y[k]$$

3 State vector

$$\mathbf{x}[k] = \left[\, \delta\bar{\mathbf{p}}_1^T[k] \;\; \delta\bar{\mathbf{p}}_2^T[k] \;\; \delta\bar{\mathbf{p}}_3^T[k] \;\; \bar{a}_T[k] \;\; \bar{a}'_T[k] \;\; \bar{a}_N[k] \;\; \bar{a}'_N[k] \;\; \theta_x[k] \;\; \theta_y[k] \,\right]^T \qquad (15 \times 1,\ d = 2)$$

4 Estimator

$$\hat{\mathbf{V}}[k] = \mathbf{R}_y(\hat{\theta}_y[k])\, \mathbf{R}_x(\hat{\theta}_x[k]) = [\, \hat{\mathbf{u}} \;\; \hat{\mathbf{v}} \;\; \hat{\mathbf{w}} \,]$$

$$\hat{\bar{\mathbf{A}}}[k] = \hat{\mathbf{V}}[k]\, \mathrm{diag}\big(\hat{\bar{a}}_T[k],\ \hat{\bar{a}}_T[k],\ \hat{\bar{a}}_N[k]\big)\, \hat{\mathbf{V}}^T[k] = \hat{\bar{a}}_T[k]\, \mathbf{I} + \big(\hat{\bar{a}}_N[k] - \hat{\bar{a}}_T[k]\big)\, \hat{\mathbf{w}}\, \hat{\mathbf{w}}^T$$

$$\hat{\mathbf{M}}[k] = \Delta\bar{\mathbf{p}}_d[k] + (1 - \lambda_c)\, \delta\hat{\bar{\mathbf{p}}}_3[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_1[k+1] = \delta\hat{\bar{\mathbf{p}}}_2[k] + \mathbf{L}_1[k]\, \mathbf{e}_y[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_2[k+1] = \delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_2[k]\, \mathbf{e}_y[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_3[k+1] = \lambda_c\, \delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_3[k]\, \mathbf{e}_y[k]$$

$$\hat{\bar{a}}_T[k+1] = \hat{\bar{a}}_T[k] + \hat{\bar{a}}'_T[k]\, \hat{\mathbf{w}}^T \hat{\mathbf{M}}[k] + \mathbf{L}_4[k]\, \mathbf{e}_y[k]$$

$$\hat{\bar{a}}'_T[k+1] = \hat{\bar{a}}'_T[k] + \mathbf{L}_5[k]\, \mathbf{e}_y[k]$$

$$\hat{\bar{a}}_N[k+1] = \hat{\bar{a}}_N[k] + \hat{\bar{a}}'_N[k]\, \hat{\mathbf{w}}^T \hat{\mathbf{M}}[k] + \mathbf{L}_6[k]\, \mathbf{e}_y[k]$$

$$\hat{\bar{a}}'_N[k+1] = \hat{\bar{a}}'_N[k] + \mathbf{L}_7[k]\, \mathbf{e}_y[k]$$

$$\hat{\theta}_x[k+1] = \hat{\theta}_x[k] + \mathbf{L}_8[k]\, \mathbf{e}_y[k]$$

$$\hat{\theta}_y[k+1] = \hat{\theta}_y[k] + \mathbf{L}_9[k]\, \mathbf{e}_y[k]$$

$\mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3$: 3×7; $\mathbf{L}_4, \ldots, \mathbf{L}_9$: 1×7; $\mathbf{e}_y = \mathbf{y} - \hat{\mathbf{y}}$ (Section 6)

5 Dynamics of estimation errors

$e_{(\cdot)} = (\cdot) - (\hat{\cdot})$; $\mathbf{e}_3 = \delta\bar{\mathbf{p}}_3 - \delta\hat{\bar{\mathbf{p}}}_3$; first order in the errors

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]$$
$$\mathbf{q}[k] = \begin{bmatrix} \mathbf{0}^T & \mathbf{0}^T & -\boldsymbol{\varepsilon}_p^T[k] & \hat{\bar{a}}'_T\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p[k] & w_{\bar{a}'_T}[k] & \hat{\bar{a}}'_N\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p[k] & w_{\bar{a}'_N}[k] & 0 & 0 \end{bmatrix}^T$$

(a) Each factor written as its estimate plus its error

$$\mathbf{w} - \hat{\mathbf{w}} = -\hat{\mathbf{v}}\,e_{\theta_x} + \cos\hat{\theta}_x\,\hat{\mathbf{u}}\,e_{\theta_y}$$
$$\mathbf{M} - \hat{\mathbf{M}} = (1-\lambda_c)\,\mathbf{e}_3$$
$$\begin{aligned}\bar{\mathbf{A}} - \hat{\bar{\mathbf{A}}} = {} & \left(\hat{\mathbf{u}}\,\hat{\mathbf{u}}^T + \hat{\mathbf{v}}\,\hat{\mathbf{v}}^T\right) e_{\bar{a}_T} + \hat{\mathbf{w}}\,\hat{\mathbf{w}}^T\,e_{\bar{a}_N} \\ & - (\hat{\bar{a}}_N - \hat{\bar{a}}_T)\left(\hat{\mathbf{v}}\,\hat{\mathbf{w}}^T + \hat{\mathbf{w}}\,\hat{\mathbf{v}}^T\right) e_{\theta_x} + (\hat{\bar{a}}_N - \hat{\bar{a}}_T)\,\cos\hat{\theta}_x\left(\hat{\mathbf{u}}\,\hat{\mathbf{w}}^T + \hat{\mathbf{w}}\,\hat{\mathbf{u}}^T\right) e_{\theta_y}\end{aligned}$$

(b) Motion along the surface normal: true minus estimated

$$\begin{aligned}&\mathbf{w}^T\left(\mathbf{M} + \left(\bar{\mathbf{A}} - \hat{\bar{\mathbf{A}}}\right)\mathbf{F}_{dp} + \boldsymbol{\varepsilon}_p\right) - \hat{\mathbf{w}}^T\hat{\mathbf{M}} \\ &= (1-\lambda_c)\,\hat{\mathbf{w}}^T\mathbf{e}_3 + \hat{\mathbf{w}}^T\mathbf{F}_{dp}\,e_{\bar{a}_N} + \left((\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\hat{\mathbf{v}}^T\mathbf{F}_{dp} - \hat{\mathbf{v}}^T\hat{\mathbf{M}}\right) e_{\theta_x} \\ &\quad + \cos\hat{\theta}_x\left(\hat{\mathbf{u}}^T\hat{\mathbf{M}} - (\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\hat{\mathbf{u}}^T\mathbf{F}_{dp}\right) e_{\theta_y} + \hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p\end{aligned}$$

(c) Errors at $k+1$

$$\mathbf{e}_1[k+1] = \mathbf{e}_2[k]$$
$$\mathbf{e}_2[k+1] = \mathbf{e}_3[k]$$
$$\begin{aligned}\mathbf{e}_3[k+1] = {} & \lambda_c\,\mathbf{e}_3[k] - \left(\hat{\mathbf{u}}\,\hat{\mathbf{u}}^T + \hat{\mathbf{v}}\,\hat{\mathbf{v}}^T\right)\mathbf{F}_{dp}[k]\,e_{\bar{a}_T}[k] - \hat{\mathbf{w}}\,\hat{\mathbf{w}}^T\mathbf{F}_{dp}[k]\,e_{\bar{a}_N}[k] \\ & + (\hat{\bar{a}}_N - \hat{\bar{a}}_T)\left(\hat{\mathbf{v}}\,\hat{\mathbf{w}}^T + \hat{\mathbf{w}}\,\hat{\mathbf{v}}^T\right)\mathbf{F}_{dp}[k]\,e_{\theta_x}[k] - (\hat{\bar{a}}_N - \hat{\bar{a}}_T)\,\cos\hat{\theta}_x\left(\hat{\mathbf{u}}\,\hat{\mathbf{w}}^T + \hat{\mathbf{w}}\,\hat{\mathbf{u}}^T\right)\mathbf{F}_{dp}[k]\,e_{\theta_y}[k] - \boldsymbol{\varepsilon}_p[k]\end{aligned}$$
$$\begin{aligned}e_{\bar{a}_T}[k+1] = {} & \hat{\bar{a}}'_T\,(1-\lambda_c)\,\hat{\mathbf{w}}^T\mathbf{e}_3[k] + e_{\bar{a}_T}[k] + \hat{\mathbf{w}}^T\hat{\mathbf{M}}\,e_{\bar{a}'_T}[k] + \hat{\bar{a}}'_T\,\hat{\mathbf{w}}^T\mathbf{F}_{dp}\,e_{\bar{a}_N}[k] \\ & + \hat{\bar{a}}'_T\left((\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\hat{\mathbf{v}}^T\mathbf{F}_{dp} - \hat{\mathbf{v}}^T\hat{\mathbf{M}}\right) e_{\theta_x}[k] + \hat{\bar{a}}'_T\,\cos\hat{\theta}_x\left(\hat{\mathbf{u}}^T\hat{\mathbf{M}} - (\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\hat{\mathbf{u}}^T\mathbf{F}_{dp}\right) e_{\theta_y}[k] \\ & + \hat{\bar{a}}'_T\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p[k]\end{aligned}$$
$$e_{\bar{a}'_T}[k+1] = e_{\bar{a}'_T}[k] + w_{\bar{a}'_T}[k]$$
$$\begin{aligned}e_{\bar{a}_N}[k+1] = {} & \hat{\bar{a}}'_N\,(1-\lambda_c)\,\hat{\mathbf{w}}^T\mathbf{e}_3[k] + \left(1 + \hat{\bar{a}}'_N\,\hat{\mathbf{w}}^T\mathbf{F}_{dp}\right) e_{\bar{a}_N}[k] + \hat{\mathbf{w}}^T\hat{\mathbf{M}}\,e_{\bar{a}'_N}[k] \\ & + \hat{\bar{a}}'_N\left((\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\hat{\mathbf{v}}^T\mathbf{F}_{dp} - \hat{\mathbf{v}}^T\hat{\mathbf{M}}\right) e_{\theta_x}[k] + \hat{\bar{a}}'_N\,\cos\hat{\theta}_x\left(\hat{\mathbf{u}}^T\hat{\mathbf{M}} - (\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\hat{\mathbf{u}}^T\mathbf{F}_{dp}\right) e_{\theta_y}[k] \\ & + \hat{\bar{a}}'_N\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p[k]\end{aligned}$$
$$e_{\bar{a}'_N}[k+1] = e_{\bar{a}'_N}[k] + w_{\bar{a}'_N}[k]$$
$$e_{\theta_x}[k+1] = e_{\theta_x}[k]$$
$$e_{\theta_y}[k+1] = e_{\theta_y}[k]$$

6 Process outputs and output matrix

Real-time calculation of process outputs (three axes)

$$
\begin{aligned}
\overline{\delta\bar{\mathbf{p}}_m}[k+1] &= a_{\mathrm{pd}}\,\delta\bar{\mathbf{p}}_m[k] + (1-a_{\mathrm{pd}})\,\overline{\delta\bar{\mathbf{p}}_m}[k] \\
\delta\bar{\mathbf{p}}_r[k] &= \delta\bar{\mathbf{p}}_m[k] - \overline{\delta\bar{\mathbf{p}}_m}[k] \\
\mathrm{E}\big[\delta\bar{\mathbf{p}}_r\,\delta\bar{\mathbf{p}}_r^T\big] &= C_T\,\kappa_T\,\bar{\mathbf{A}} + C_n\,\boldsymbol{\Sigma}_n, \qquad \boldsymbol{\Sigma}_n = \mathrm{diag}\big(\sigma_{n,x}^2,\ \sigma_{n,y}^2,\ \sigma_{n,z}^2\big) \\
{}^{v}\delta\bar{\mathbf{p}}_r[k] &= \hat{\mathbf{V}}^T[k]\,\delta\bar{\mathbf{p}}_r[k] \\
{}^{v}\boldsymbol{\Sigma}_n[k] &= \hat{\mathbf{V}}^T[k]\,\boldsymbol{\Sigma}_n\,\hat{\mathbf{V}}[k] \\
{}^{v}\bar{\mathbf{A}}_m[k] &= \frac{{}^{v}\delta\bar{\mathbf{p}}_r[k]\ {}^{v}\delta\bar{\mathbf{p}}_r^T[k] - C_n\ {}^{v}\boldsymbol{\Sigma}_n[k]}{C_T\,\kappa_T} \\
\bar{a}_{Tm}[k] &= \frac{\big[{}^{v}\bar{\mathbf{A}}_m[k]\big]_{uu} + \big[{}^{v}\bar{\mathbf{A}}_m[k]\big]_{vv}}{2} \\
\bar{a}_{Nm}[k] &= \big[{}^{v}\bar{\mathbf{A}}_m[k]\big]_{ww} \\
\theta_{xm}[k] &= \hat{\theta}_x[k] + \frac{\big[{}^{v}\bar{\mathbf{A}}_m[k]\big]_{vw}}{\hat{\bar{a}}_T[k] - \hat{\bar{a}}_N[k]} \\
\theta_{ym}[k] &= \hat{\theta}_y[k] - \frac{\big[{}^{v}\bar{\mathbf{A}}_m[k]\big]_{uw}}{\big(\hat{\bar{a}}_T[k] - \hat{\bar{a}}_N[k]\big)\,\cos\hat{\theta}_x[k]}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{E}\big[{}^{v}\bar{\mathbf{A}}_m\big] &= \hat{\mathbf{V}}^T\,\bar{\mathbf{A}}\,\hat{\mathbf{V}} \\
\mathrm{E}\big[{}^{v}\bar{\mathbf{A}}_m\big]_{uu} &\approx \bar{a}_T \\
\mathrm{E}\big[{}^{v}\bar{\mathbf{A}}_m\big]_{vv} &\approx \bar{a}_T \\
\mathrm{E}\big[{}^{v}\bar{\mathbf{A}}_m\big]_{ww} &\approx \bar{a}_N \\
\mathrm{E}\big[{}^{v}\bar{\mathbf{A}}_m\big]_{vw} &\approx (\bar{a}_T - \bar{a}_N)\,e_{\theta_x} \\
\mathrm{E}\big[{}^{v}\bar{\mathbf{A}}_m\big]_{uw} &\approx -(\bar{a}_T - \bar{a}_N)\,\cos\hat{\theta}_x\,e_{\theta_y}
\end{aligned}
$$

Measurement, estimator output and output matrix

$$
\mathbf{y}[k] = \begin{bmatrix} \delta\bar{\mathbf{p}}_m[k] \\ \bar{a}_{Tm}[k] \\ \bar{a}_{Nm}[k] \\ \theta_{xm}[k] \\ \theta_{ym}[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{\mathbf{p}}}_1[k] \\ \hat{\bar{a}}_T[k] \\ \hat{\bar{a}}_N[k] \\ \hat{\theta}_x[k] \\ \hat{\theta}_y[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 1 & 0 & 0 & 0 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 1 & 0 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

7 Process noise and measurement noise

$$\mathbf{Q}_{33}[k] = \kappa_T \left(\hat{\bar{\mathbf{A}}}[k] + (1-\lambda_c)^2 \sum_{i=1}^{d} \hat{\bar{\mathbf{A}}}[k-i] \right) + (1-\lambda_c)^2 \, \boldsymbol{\Sigma}_n$$

$$q_{\bar{a}'_T}[k] = p^2_{\text{fix}}[k] \, I_{\bar{a}'_T}[k], \qquad q_{\bar{a}'_N}[k] = p^2_{\text{fix}}[k] \, I_{\bar{a}'_N}[k]$$

$$I_{(\cdot)}[k] = \frac{1}{p_{(\cdot)}[k]} - \frac{1}{p^f_{(\cdot)}[k]}$$

$$\sqrt{p_{\text{fix}}}[k] = 0.4 \left(1.1/h_d[k]\right)^2, \qquad h_d[k] = \hat{\mathbf{w}}^T[k] \, \bar{\mathbf{p}}_d[k]$$

$$\mathbf{Q}[k] = \begin{bmatrix}
\mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \mathbf{Q}_{33} & -\hat{\bar{a}}'_T \, \mathbf{Q}_{33}\hat{\mathbf{w}} & \mathbf{0} & -\hat{\bar{a}}'_N \, \mathbf{Q}_{33}\hat{\mathbf{w}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & -\hat{\bar{a}}'_T \, \hat{\mathbf{w}}^T\mathbf{Q}_{33} & \hat{\bar{a}}'^2_T \, \hat{\mathbf{w}}^T\mathbf{Q}_{33}\hat{\mathbf{w}} & 0 & \hat{\bar{a}}'_T\hat{\bar{a}}'_N \, \hat{\mathbf{w}}^T\mathbf{Q}_{33}\hat{\mathbf{w}} & 0 & 0 & 0 \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & q_{\bar{a}'_T} & 0 & 0 & 0 & 0 \\
\mathbf{0} & \mathbf{0} & -\hat{\bar{a}}'_N \, \hat{\mathbf{w}}^T\mathbf{Q}_{33} & \hat{\bar{a}}'_T\hat{\bar{a}}'_N \, \hat{\mathbf{w}}^T\mathbf{Q}_{33}\hat{\mathbf{w}} & 0 & \hat{\bar{a}}'^2_N \, \hat{\mathbf{w}}^T\mathbf{Q}_{33}\hat{\mathbf{w}} & 0 & 0 & 0 \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & q_{\bar{a}'_N} & 0 & 0 \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\xi_T = \frac{C_n \left(\hat{\mathbf{u}}^T \boldsymbol{\Sigma}_n \, \hat{\mathbf{u}} + \hat{\mathbf{v}}^T \boldsymbol{\Sigma}_n \, \hat{\mathbf{v}} \right)}{2 \, C_T \, \kappa_T}, \qquad \xi_N = \frac{C_n \, \hat{\mathbf{w}}^T \boldsymbol{\Sigma}_n \, \hat{\mathbf{w}}}{C_T \, \kappa_T}$$

$$\mathbf{R}[k] = \text{diag}\Big(\boldsymbol{\Sigma}_n, \;\; \text{IF} \left(\hat{\bar{a}}_T + \xi_T\right)^2, \;\; 2 \, \text{IF} \left(\hat{\bar{a}}_N + \xi_N\right)^2, \;\; \frac{\text{IF} \left(\hat{\bar{a}}_T + \xi_T\right)\left(\hat{\bar{a}}_N + \xi_N\right)}{\left(\hat{\bar{a}}_T - \hat{\bar{a}}_N\right)^2}, \;\; \frac{\text{IF} \left(\hat{\bar{a}}_T + \xi_T\right)\left(\hat{\bar{a}}_N + \xi_N\right)}{\left(\hat{\bar{a}}_T - \hat{\bar{a}}_N\right)^2 \cos^2 \hat{\theta}_x} \Big)$$

Estimator feedback matrix and error covariance

$$\mathbf{L}[k] = \mathbf{P}^f[k] \, \mathbf{H}^T \left(\mathbf{H} \, \mathbf{P}^f[k] \, \mathbf{H}^T + \mathbf{R}[k] \right)^{-1}$$

$$\mathbf{P}[k] = \left(\mathbf{I} - \mathbf{L}[k] \, \mathbf{H} \right) \mathbf{P}^f[k]$$

$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k] \, \mathbf{P}[k] \, \mathbf{F}_e^T[k] + \mathbf{Q}[k]$$

Seed and prior

$$\hat{\theta}_x[0] = 0, \qquad \hat{\theta}_y[0] = 0$$

$$h_d[0] = \hat{\mathbf{w}}^T[0] \, \bar{\mathbf{p}}_d[0]$$

$$\hat{\bar{a}}_T[0] = \frac{1}{c_{\parallel}(h_d[0])}, \qquad \hat{\bar{a}}_N[0] = \frac{1}{c_{\perp}(h_d[0])}$$

$$\hat{\bar{a}}'_T[0] = \frac{9/16}{h_d[0]^2}, \qquad \hat{\bar{a}}'_N[0] = \frac{1}{h_d[0]^2}$$

$$\mathbf{P}[0] = \text{diag}\big(\mathbf{P}_\delta[0], \; p_{\bar{a}_T}[0], \; p_{\bar{a}'_T}[0], \; p_{\bar{a}_N}[0], \; p_{\bar{a}'_N}[0], \; p_\theta[0], \; p_\theta[0] \big)$$